

Clustering

Material

<https://youtu.be/kqcvqqZp3Jo?si=T1lHw0z-4wv6BvF->

- Evaluation of clustering methods

Exercises

Load the seu dataset you have created earlier today:

The method implemented in Seurat first constructs a SNN graph based on the euclidean distance in PCA space, and refine the edge weights between any two cells based on the shared overlap in their local neighborhoods (Jaccard similarity). This step is performed using the `FindNeighbors()` function, and takes as input the previously defined dimensionality of the dataset.

Code starts here:

```
seu <- Seurat::FindNeighbors(seu, dims = 1:25, reduction = "integrated.cca")
```

To cluster the cells, Seurat next implements modularity optimization techniques such as the Louvain algorithm (default) or SLM [SLM, Blondel et al., Journal of Statistical Mechanics], to iteratively group cells together, with the goal of optimizing the standard modularity function. The `FindClusters()` function implements this procedure, and contains a resolution parameter that sets the ‘granularity’ of the downstream clustering, with increased values leading to a greater number of clusters.

Code starts here:

```
seu <- Seurat::FindClusters(seu, resolution = seq(0.1, 0.8, by=0.1))
```

Cluster id of each cell is added to the metadata object, as a new column for each resolution tested:

Code starts here:

```
head(seu@meta.data)
```

	orig.ident	nCount_RNA	nFeature_RNA	percent.mito
PBMMC-1_AAACCTGCAGACGCAA-1	PBMMC-1	2401	909	2.540608
PBMMC-1_AAACCTGTCATCACCC-1	PBMMC-1	3532	760	5.181200
PBMMC-1_AAAGATGCATAAAGGT-1	PBMMC-1	3972	1215	4.934542
PBMMC-1_AAAGCAAAGCAGCGTA-1	PBMMC-1	3569	894	3.250210
PBMMC-1_AAAGCAACAATAACGA-1	PBMMC-1	2982	730	3.688799
PBMMC-1_AAAGCAACATCAGTCA-1	PBMMC-1	22284	3108	3.181655

```
percent.ribo percent.globin nCount_SCT  
nFeature_SCT
```

PBMMC-1_AAACCTGCAGACGCAA-1 886	28.65473	0.1665973	3577
PBMMC-1_AAACCTGTCATCACCC-1 747	55.03964	0.1981880	3909
PBMMC-1_AAAGATGCATAAAGGT-1 1192	30.43807	0.3776435	3964
PBMMC-1_AAAGCAAAGCAGCGTA-1 873	55.02942	0.3642477	3883
PBMMC-1_AAAGCAACAATAACGA-1 711	54.49363	0.1006036	3795
PBMMC-1_AAAGCAACATCAGTCA-1 950	23.40693	36.9682283	3967
	RNA_snn_res.0.1	RNA_snn_res.0.2	RNA_snn_res.0.3
PBMMC-1_AAACCTGCAGACGCAA-1	5	7	7
PBMMC-1_AAACCTGTCATCACCC-1	0	0	0
PBMMC-1_AAAGATGCATAAAGGT-1	2	3	3
PBMMC-1_AAAGCAAAGCAGCGTA-1	0	0	0
PBMMC-1_AAAGCAACAATAACGA-1	0	0	0
PBMMC-1_AAAGCAACATCAGTCA-1	4	2	2
	RNA_snn_res.0.4	RNA_snn_res.0.5	RNA_snn_res.0.6
PBMMC-1_AAACCTGCAGACGCAA-1	7	7	9
PBMMC-1_AAACCTGTCATCACCC-1	0	0	0
PBMMC-1_AAAGATGCATAAAGGT-1	2	3	3
PBMMC-1_AAAGCAAAGCAGCGTA-1	0	0	0
PBMMC-1_AAAGCAACAATAACGA-1	0	0	0
PBMMC-1_AAAGCAACATCAGTCA-1	5	5	5
	RNA_snn_res.0.7	RNA_snn_res.0.8	seurat_clusters
PBMMC-1_AAACCTGCAGACGCAA-1	9	8	8
PBMMC-1_AAACCTGTCATCACCC-1	0	0	0
PBMMC-1_AAAGATGCATAAAGGT-1	2	2	2
PBMMC-1_AAAGCAAAGCAGCGTA-1	0	0	0
PBMMC-1_AAAGCAACAATAACGA-1	0	0	0
PBMMC-1_AAAGCAACATCAGTCA-1	5	5	5

To view how clusters sub-divide at increasing resolution:

Code starts here:

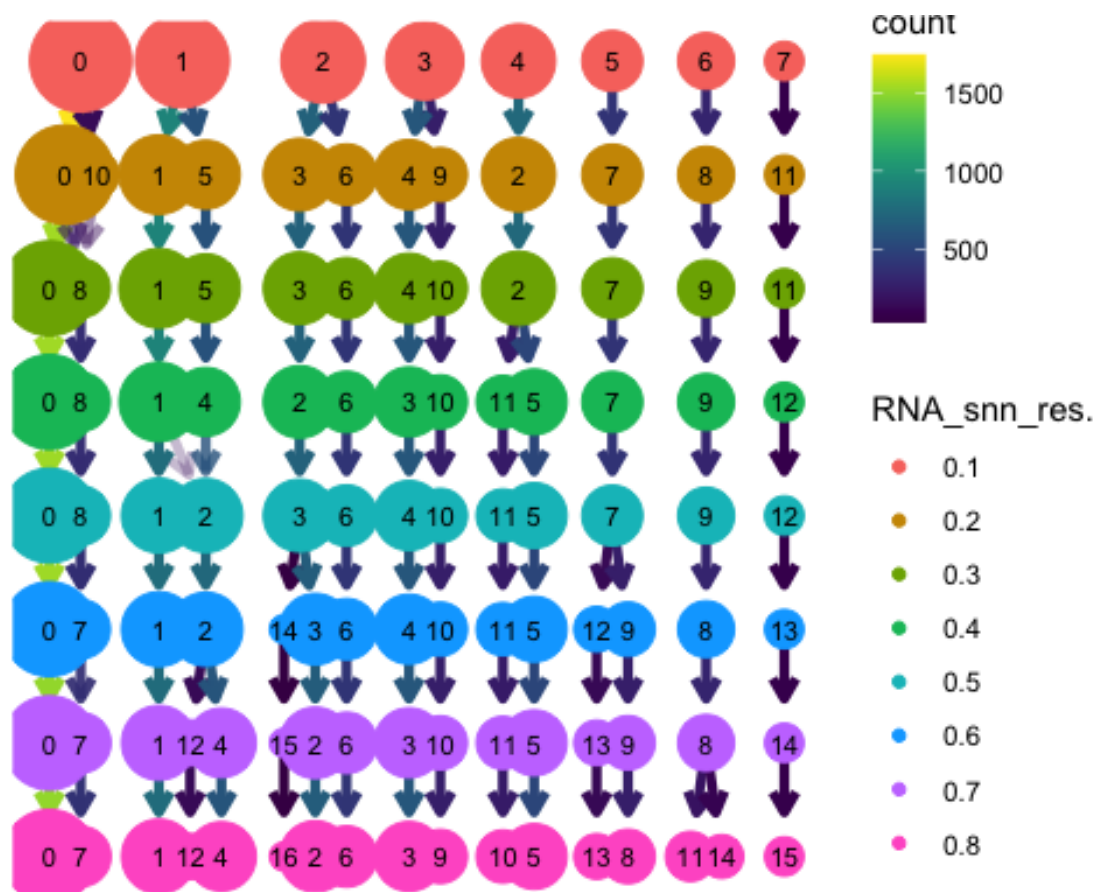
```
library(clustree)

Loading required package: ggraph

Loading required package: ggplot2

Warning: package 'ggplot2' was built under R version 4.3.3

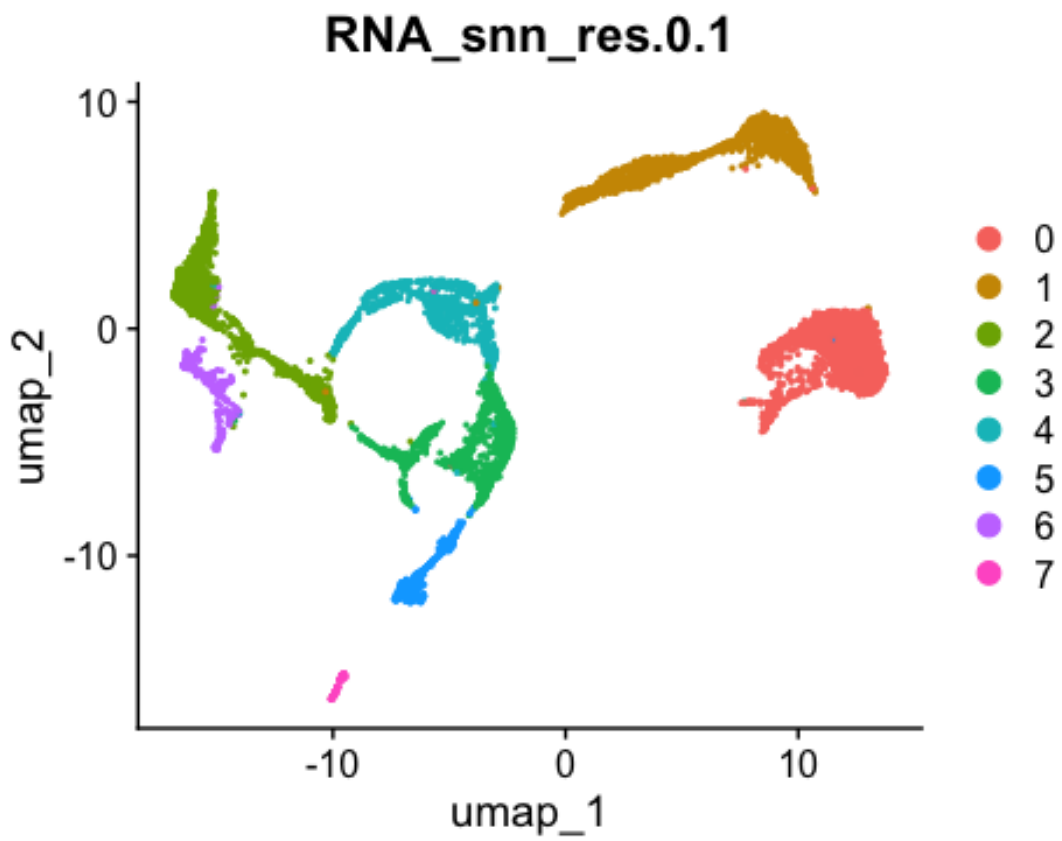
clustree::clustree(seu@meta.data[,grep("RNA_snn_res",
colnames(seu@meta.data))],
                  prefix = "RNA_snn_res.")
```



You can view the UMAP coloring each cell according to a cluster id like this:

Code starts here:

```
Seurat::DimPlot(seu, group.by = "RNA_snn_res.0.1")
```



⚠ Exercise

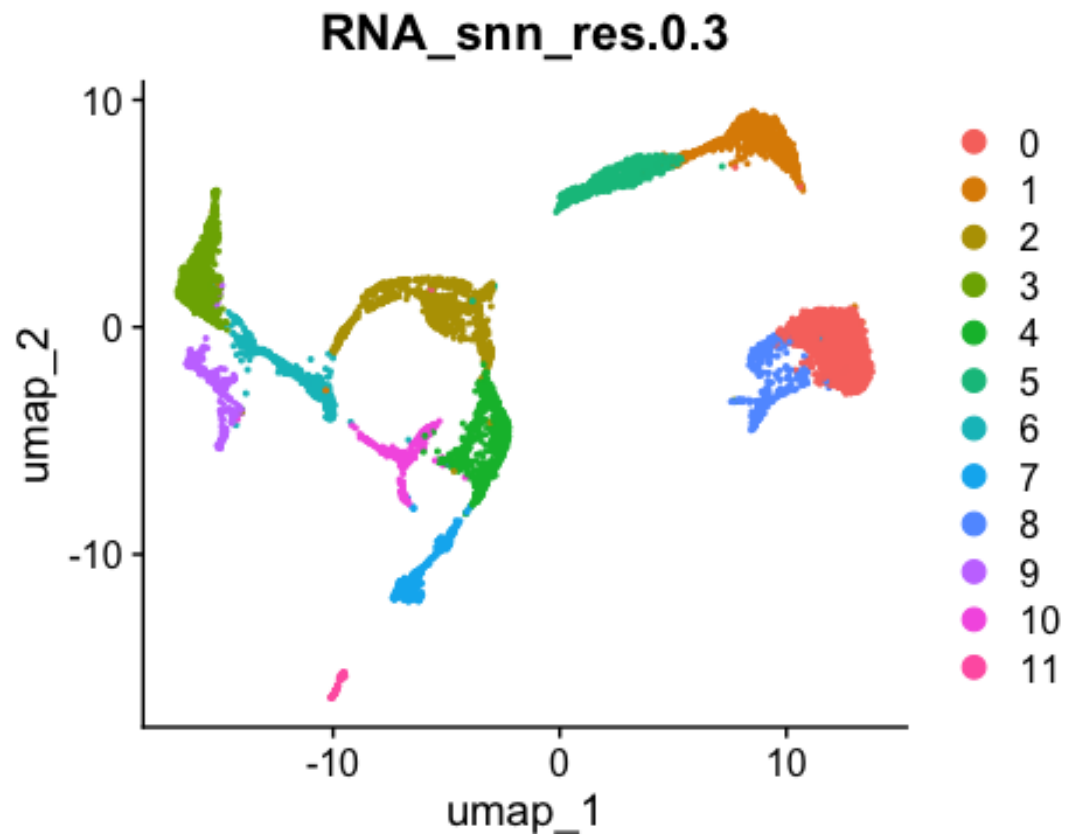
Visualise clustering based on a few more resolutions. Taking the clustering and the UMAP plots into account what do you consider as a good resolution to perform the clustering?

💡 Answer

Of course, there is no 'optimal' resolution, but based on resolution of 0.3, the tree stays relatively stable for a few resolution steps, and it seems that clustering fits the UMAP well:

Code starts here:

```
Seurat::DimPlot(seu, group.by = "RNA_snn_res.0.3")
```



⚠ Exercise

When do the number of neighbors need to be changed? How does changing the method of clustering in FindClusters affect the output? Which parameter should be changed?

💡 Answer

As FindClusters is an unsupervised clustering method on the PCA data and UMAP is a good summary of the PCA dimension selected, clusters and UMAP plot should go along. If one has reasons to change the number of neighbors in the UMAP function, here the same parameter should be adapted.

The method can be changed with algorithm = 2,3 or 4